

Tutorial 9 (Dynamics Fundamental)

Ignore the parts on using computer programs to get the numerical solutions

- Problem 10-1

Additional information: densities of the brass hammer head and the wood handle are respectively 0.31 and $0.9 \times 0.036 \text{ lbf/in}^3$; density of water = 0.036 lbf/in^3 .

- Problem 10-4
- Problem 10-22 (Solutions for Problem 9-35 in Tutorial 8 are needed)
- Problem 10-24

*†10-1 The mallet shown in Figure 10-2 has the following specifications: The steel head has a 1.25-in diameter and is 3.5 in tall; the wood handle is 1.5-in diameter and 10 in long and necks down to 3/4 in wide where it enters the head. Find the location of its composite CG , and its moment of inertia and radius of gyration about axis ZZ . Assume the wood has a density equal to 0.9 times that of water.

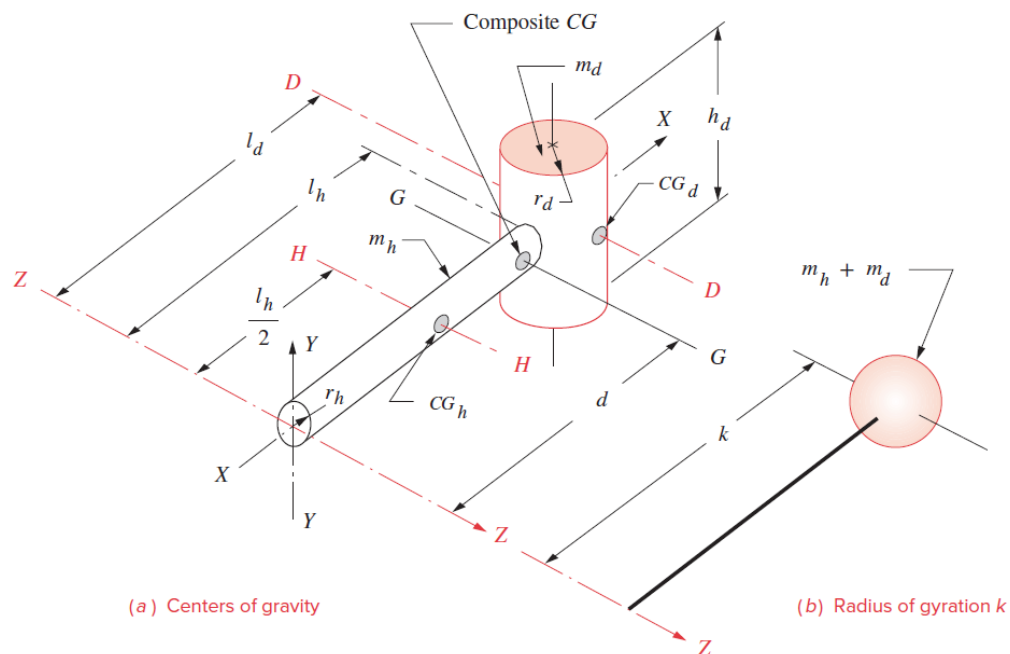


FIGURE 10-2

*†10-4 Set up these equations in matrix form. Use program MATRIX, *Mathcad*, Matlab, or a calculator which has matrix math capability to solve them.

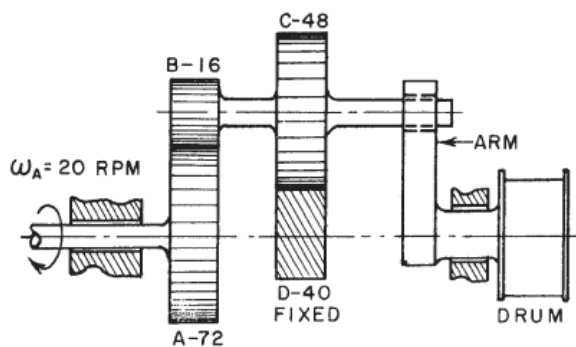
a.

$$\begin{array}{rrrrr} -5x & -2y & +12z & -w & = -9 \\ x & +3y & -2z & +4w & = 10 \\ -x & -y & +z & & = -7 \\ 3x & -3y & +7z & +9w & = -6 \end{array}$$

b.

$$\begin{array}{rrrrr} 3x & -5y & +17z & -5w & = -5 \\ -2x & +9y & -14z & +6w & = 22 \\ -x & -y & & -2w & = 13 \\ 4x & -7y & +8z & +4w & = -9 \end{array}$$

†10-22 What is the effective inertia of a load applied at the drum of Figure P9-5a as reflected back to gear A?



(a)

FIGURE P9-5

†10-24 Refer to Figure 10-10. Given $a = 100$ mm, $b = 150$ mm, $k_A = 2000$ N/m, and $m_B = 2$ kg, find the equivalent mass at point A and the equivalent spring at point B.

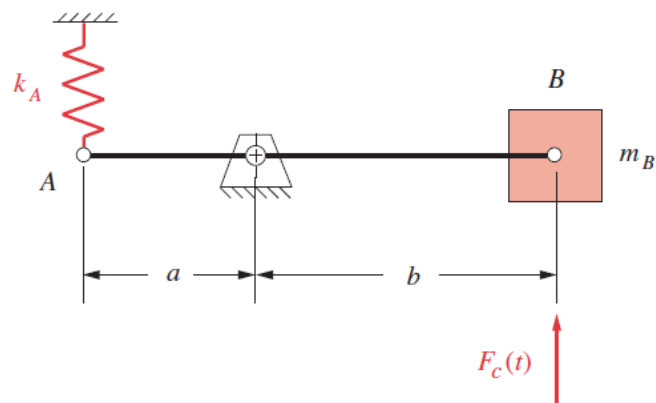


FIGURE 10-10